



Supermarket Sales Analysis



Objective

To analyze supermarket sales data using statistical and visual methods and extract key business insights.

Tools Used: Python, Pandas, Seaborn, Matplotlib

```
In [2]: pip install matplotlib seaborn pandas
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: matplotlib in c:\users\anjul\appdata\roaming\python\python312\site-packages (3.10.1)
Requirement already satisfied: seaborn in c:\users\anjul\appdata\roaming\python\python312\site-packages (0.13.2)
Requirement already satisfied: pandas in c:\users\anjul\appdata\roaming\python\python312\site-packages (2.2.3)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\anjul\appdata\roaming\python\python312\site-packages (from matplotlib) (1.3.2)
Requirement already satisfied: cycler>=0.10 in c:\users\anjul\appdata\roaming\python\python312\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\anjul\appdata\roaming\python\python312\site-packages (from matplotlib) (4.57.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\anjul\appdata\roaming\python\python312\site-packages (from matplotlib) (1.4.8)
Requirement already satisfied: numpy>=1.23 in c:\users\anjul\appdata\roaming\python\python312\site-packages (from matplotlib) (2.2.5)
Requirement already satisfied: packaging>=20.0 in c:\programdata\miniconda3\lib\site-packages (from matplotlib) (24.2)
Requirement already satisfied: pillow>=8 in c:\users\anjul\appdata\roaming\python\python312\site-packages (from matplotlib) (11.2.1)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\anjul\appdata\roaming\python\python312\site-packages (from matplotlib) (3.2.3)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\anjul\appdata\roaming\python\python312\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\anjul\appdata\roaming\python\python312\site-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\anjul\appdata\roaming\python\python312\site-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in c:\users\anjul\appdata\roaming\python\python312\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
```

Note: you may need to restart the kernel to use updated packages.

```
In [4]: # import libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [7]: # Set visualization styles
sns.set(style="whitegrid")
plt.rcParams["figure.figsize"] = (10, 6)
```

2. Load Dataset

Load the CSV into a DataFrame and take a peek at the first few rows.

```
In [9]: # Load the dataset
df = pd.read_csv("C:/Users/anjul/Downloads/SuperMarket_Analysis.csv")
```

3. Basic Exploration

- Check dataset info, descriptive statistics
- See distribution of key categorical variables

```
In [10]: # Basic exploration
print("Dataset Info:")
print(df.info())
```

Dataset Info:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 1000 entries, 0 to 999

Data columns (total 17 columns):

#	Column	Non-Null Count	Dtype
0	Invoice ID	1000 non-null	object
1	Branch	1000 non-null	object
2	City	1000 non-null	object
3	Customer type	1000 non-null	object
4	Gender	1000 non-null	object
5	Product line	1000 non-null	object
6	Unit price	1000 non-null	float64
7	Quantity	1000 non-null	int64
8	Tax 5%	1000 non-null	float64
9	Sales	1000 non-null	float64
10	Date	1000 non-null	object
11	Time	1000 non-null	object
12	Payment	1000 non-null	object
13	cogs	1000 non-null	float64
14	gross margin percentage	1000 non-null	float64
15	gross income	1000 non-null	float64
16	Rating	1000 non-null	float64

dtypes: float64(7), int64(1), object(9)

memory usage: 132.9+ KB

None

```
In [20]: print("\nDescriptive Statistics:")
print(df.describe())
```

Descriptive Statistics:

	Unit price	Quantity	Tax 5%	Sales	cogs \
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.000000
mean	55.672130	5.510000	15.379369	322.966749	307.58738
std	26.494628	2.923431	11.708825	245.885335	234.17651
min	10.080000	1.000000	0.508500	10.678500	10.17000
25%	32.875000	3.000000	5.924875	124.422375	118.49750
50%	55.230000	5.000000	12.088000	253.848000	241.76000
75%	77.935000	8.000000	22.445250	471.350250	448.90500
max	99.960000	10.000000	49.650000	1042.650000	993.00000

	gross margin percentage	gross income	Rating
count	1000.000000	1000.000000	1000.000000
mean	4.761905	15.379369	6.97270
std	0.000000	11.708825	1.71858
min	4.761905	0.508500	4.00000
25%	4.761905	5.924875	5.50000
50%	4.761905	12.088000	7.00000
75%	4.761905	22.445250	8.50000
max	4.761905	49.650000	10.00000

```
In [28]: print("\nValue Counts:")
print(df['Gender'].value_counts())
print(df['Payment'].value_counts())
```

Value Counts:

Gender

Female 571

Male 429

Name: count, dtype: int64

Payment

Ewallet 345

Cash 344

Credit card 311

Name: count, dtype: int64

? 4. Missing Values

Verify that there are no nulls in the dataset.

```
In [27]: # Check for Missing Values
print (df.isnull().sum())
```

```

Invoice ID      0
Branch          0
City            0
Customer type   0
Gender          0
Product line    0
Unit price      0
Quantity        0
Tax 5%          0
Sales           0
Date            0
Time            0
Payment         0
cogs            0
gross margin percentage 0
gross income    0
Rating          0
dtype: int64

```

5. Univariate Analysis

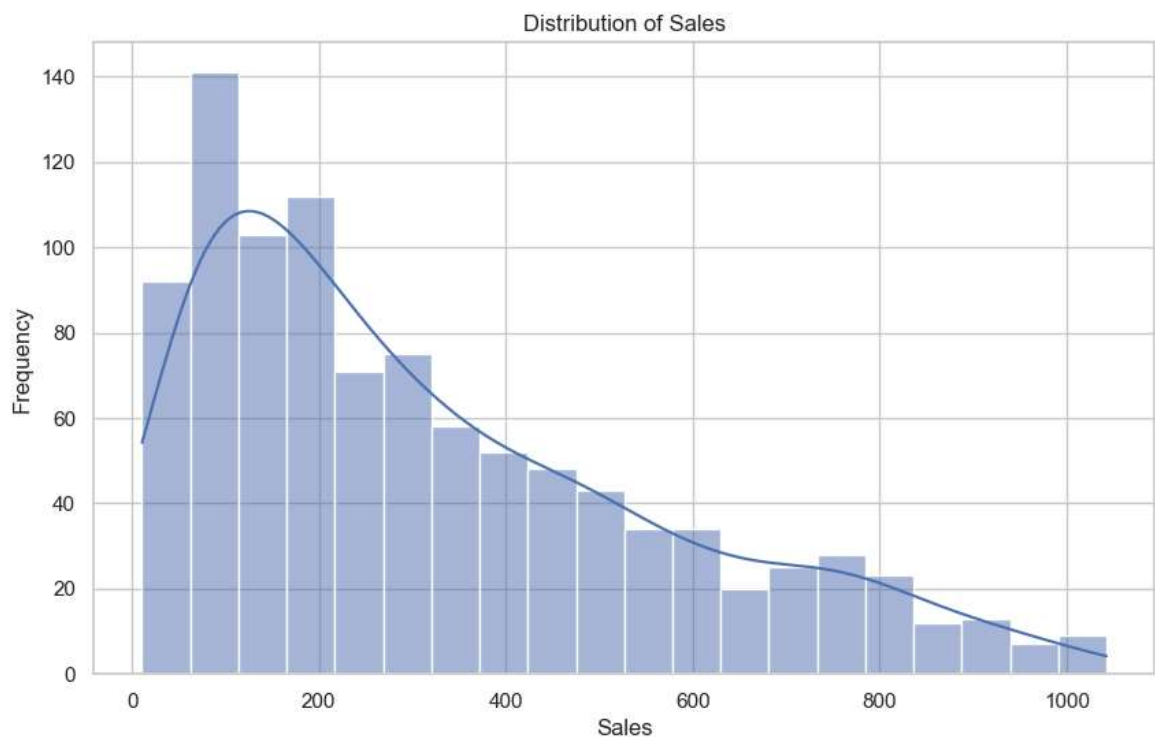
5.1 Distribution of Sales

Use a histogram to understand the spread of the `Sales` column.

```

In [11]: # Histogram of Sales
sns.histplot(df['Sales'], bins=20, kde=True)
plt.title("Distribution of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()

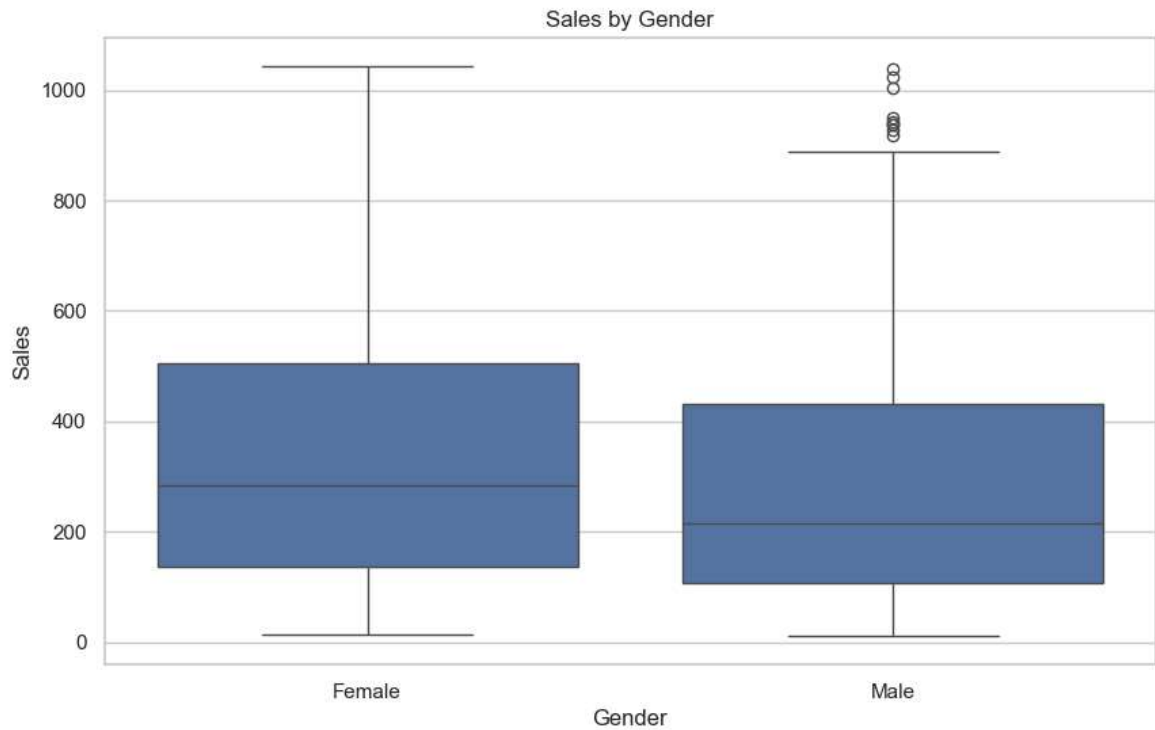
```



5.2 Sales by Gender

Use a boxplot to compare sales distributions between genders.

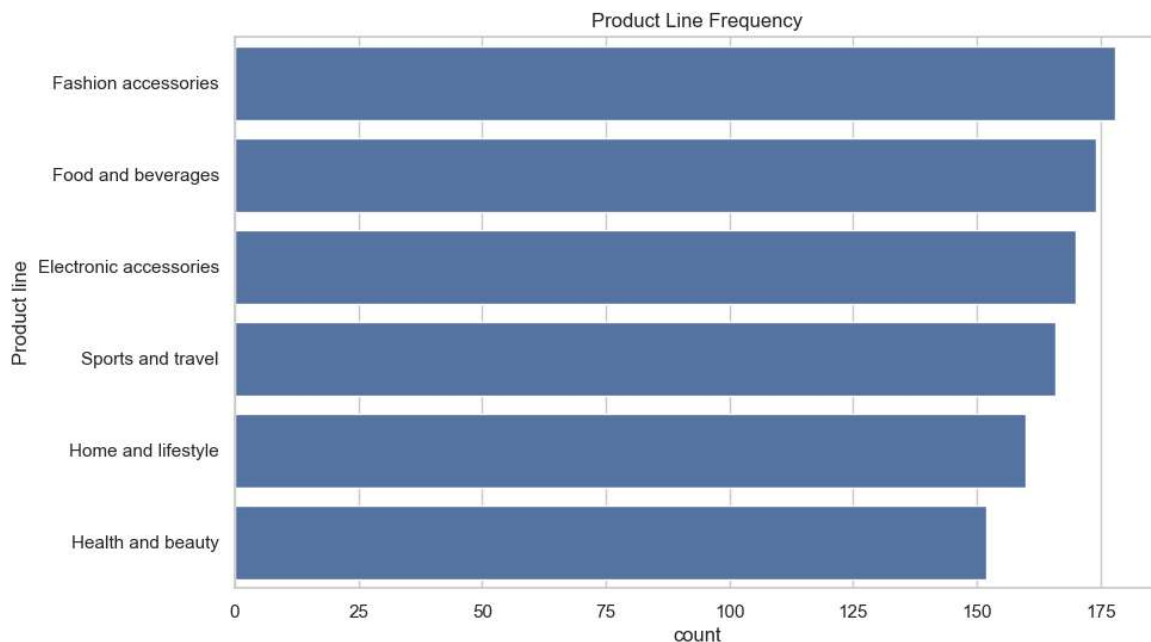
```
In [12]: # Boxplot: Sales by Gender
sns.boxplot(data=df, x='Gender', y='Sales')
plt.title("Sales by Gender")
plt.show()
```



5.3 Product Line Frequency

Count how many transactions occurred in each product line.

```
In [39]: # Countplot: Product Lines
sns.countplot(data=df, y='Product line', order=df['Product line'].value_cou
plt.title("Product Line Frequency")
plt.show()
```

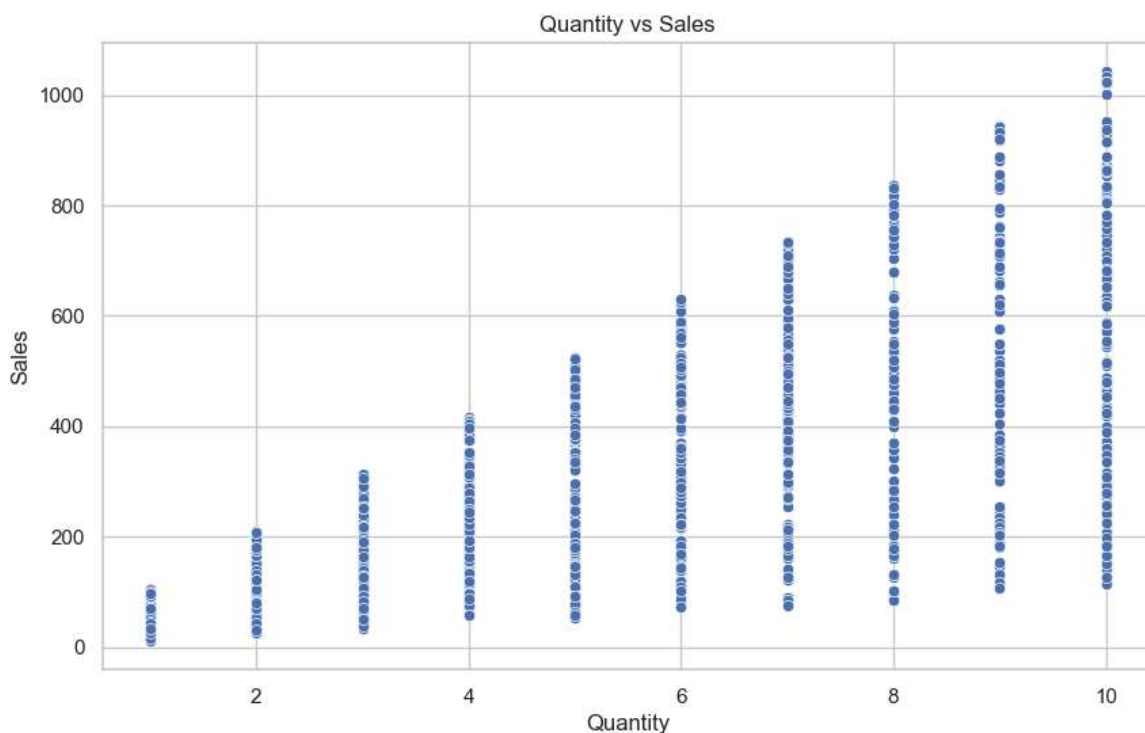


6. Bivariate Analysis

6.1 Quantity vs Sales

Scatterplot to see how quantity sold relates to total sales.

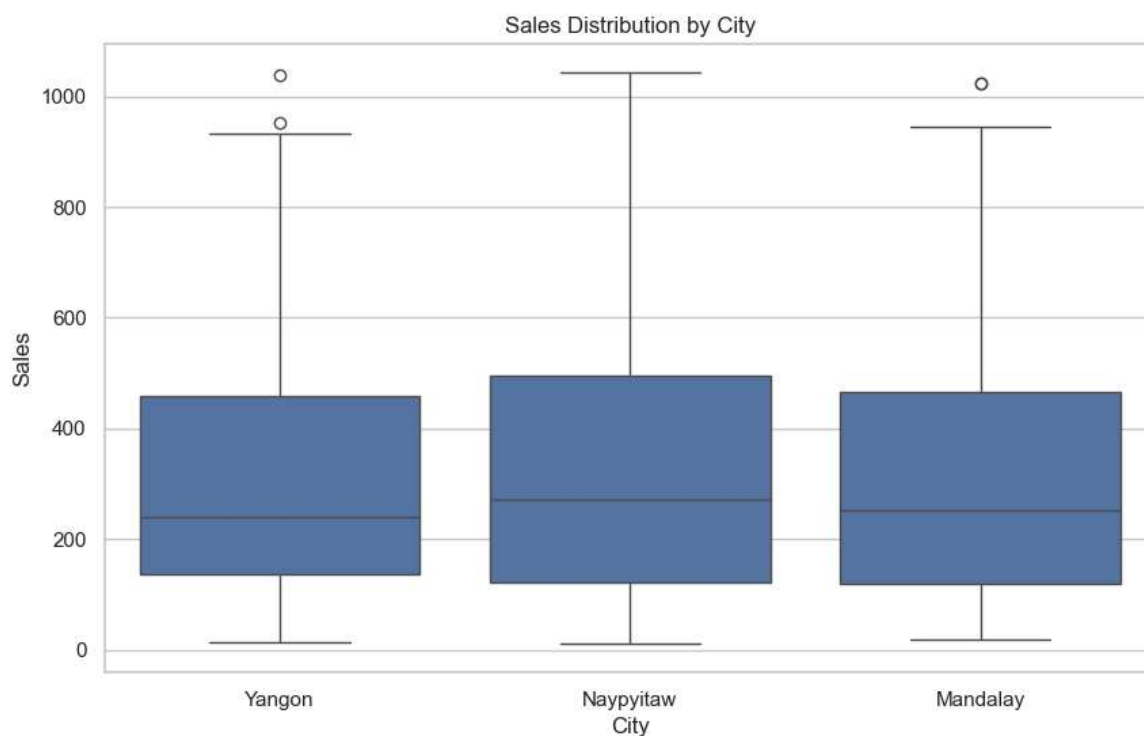
```
In [40]: # Scatterplot: Quantity vs Sales
sns.scatterplot(data=df, x='Quantity', y='Sales')
plt.title("Quantity vs Sales")
plt.show()
```



6.2 Sales by City

Boxplot to compare sales distributions across cities.

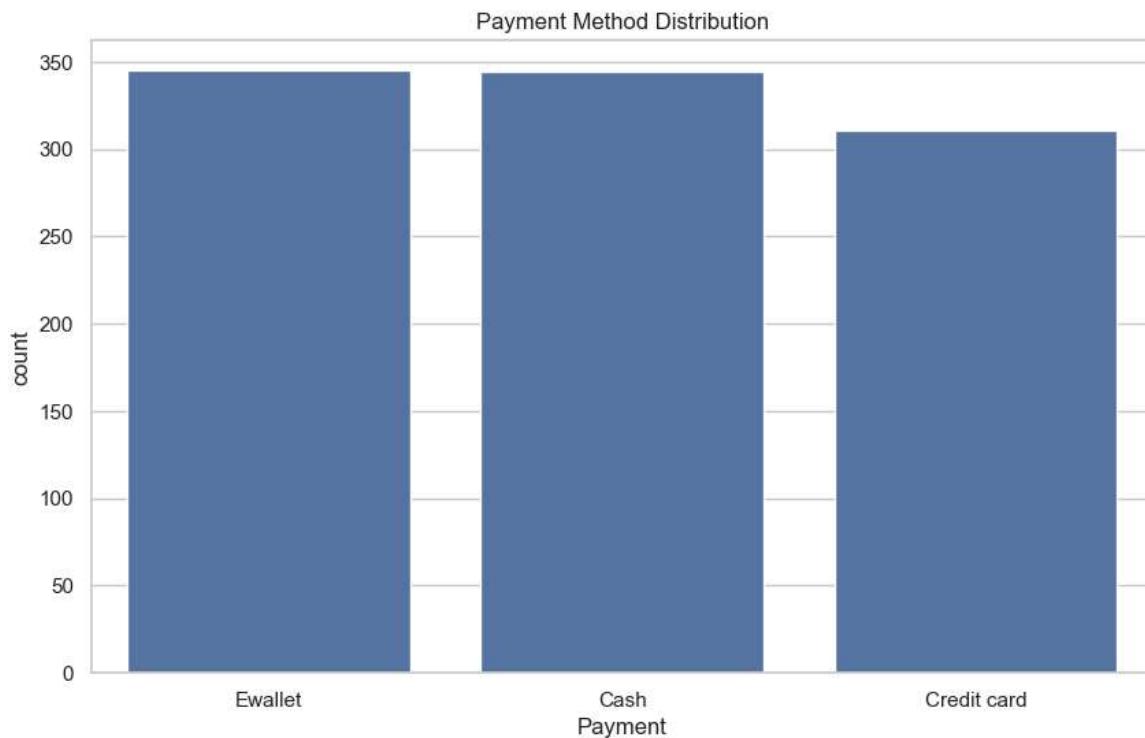
```
In [46]: # boxplot: City vs Sales
sns.boxplot(data=df, x='City', y='Sales')
plt.title("Sales Distribution by City")
plt.show()
```



6.3 Payment Method Distribution

Countplot for payment types.

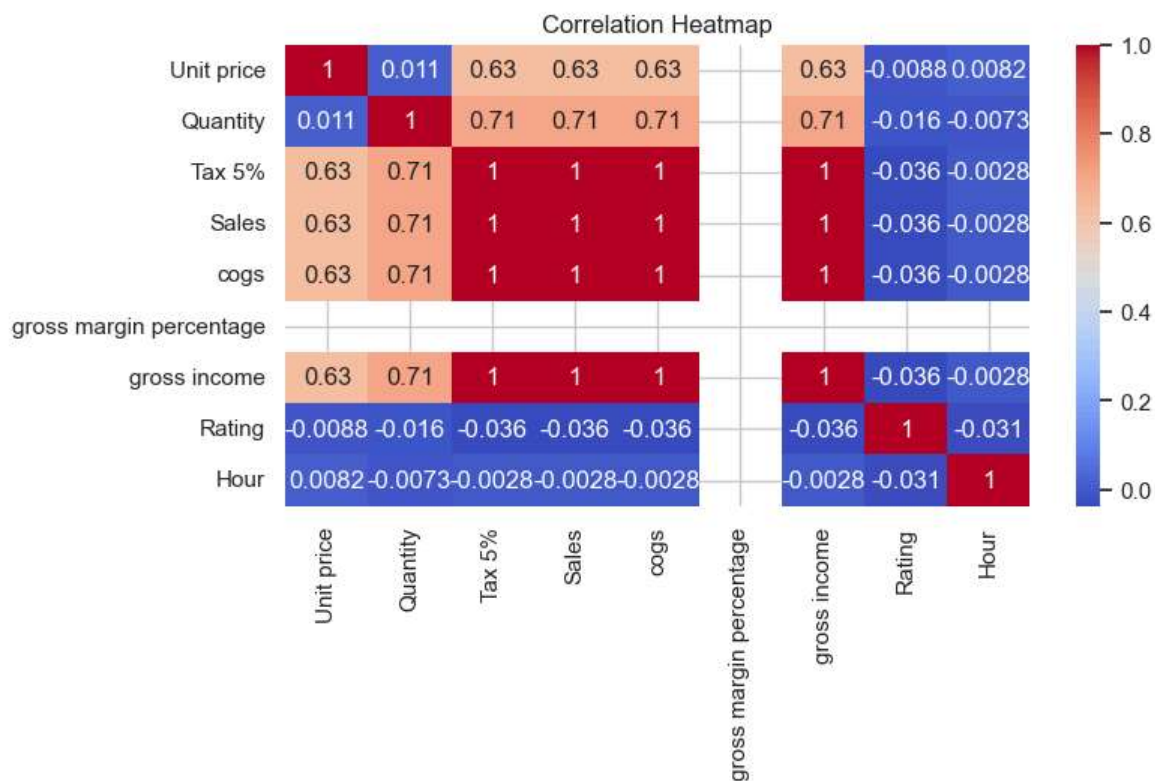
```
In [49]: # Countplot: Payment Method
sns.countplot(data=df, x='Payment')
plt.title("Payment Method Distribution")
plt.show()
```



7. Correlation Analysis

Use a heatmap to uncover relationships among numeric features.

```
In [13]: # Correlation Heatmap
plt.figure(figsize=(8,4))
sns.heatmap(df.select_dtypes(include='number').corr(), annot=True, cmap='co
plt.title("Correlation Heatmap")
plt.show()
```

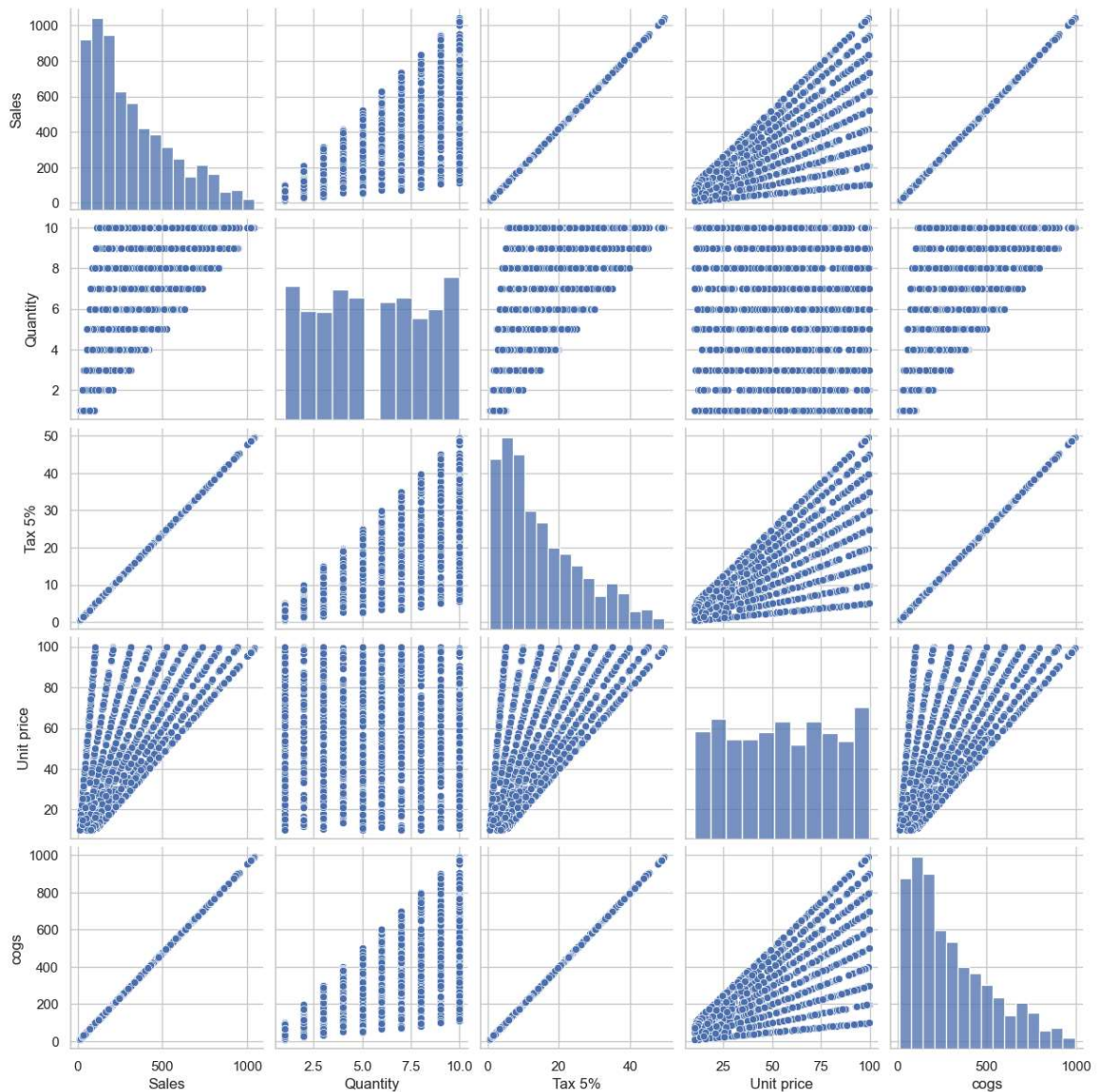




9. Pairwise Relationships

Use pairplot to visualize pairwise scatterplots and distributions.

```
In [23]: # pairplot
sns.pairplot(df[['Sales', 'Quantity', 'Tax 5%', 'Unit price', 'cogs']])
plt.show()
```



10. Summary of Findings

```
In [25]: # Summary of Findings
summary = ""
Summary of Insights:
- 'Food and beverages' and 'Health and beauty' are the most popular product
- Ewallet is the most preferred payment method.
- Male and Female customers spend similarly.
- Cities show variation in average sales, indicating different consumer beh
- Strong correlation between 'Sales', 'Tax', and 'COGS'.
```

```
print(summary)
```

Summary of Insights:

- 'Food and beverages' and 'Health and beauty' are the most popular product lines.
- Ewallet is the most preferred payment method.
- Male and Female customers spend similarly.
- Cities show variation in average sales, indicating different consumer behavior.
- Strong correlation between 'Sales', 'Tax', and 'COGS'.